

Taper-Lock® Dual Duty Pulleys



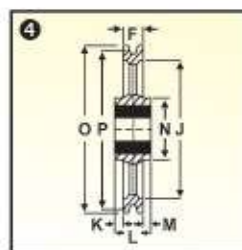
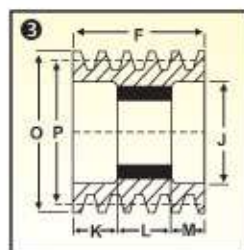
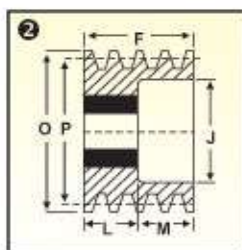
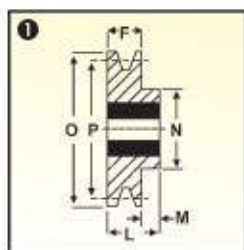
Fenner Taper-Lock® Dual Duty Pulleys are India's first standard range of metric pulleys. They eliminate reboring and keywaying problems since each pulley need not be finished according to different bore and keyway requirements. With Taper-Lock® Bushes, changes in the diameter and position of pulleys are easily made. These metric pulleys are precision machined and have dual duty grooves to perfectly match Fenner "PB" Precision Built V-Belts as well as Fenner Spacesaver Wedge Belts.

All Fenner pulleys are manufactured from superior quality cast iron in over 400 standard sizes. Also additional sizes against special requirements and special material grade can be provided. Fenner metric pulleys conform to IS and ISO groove specifications and standard sizes can transmit upto 250 kW of power at 1440 rpm with speed ratios upto 1 : 7. The products are made available with excellent pre and post - sales service through our Branches and Distributors across the country.

Fenner

POWERTRAN

Taper Lock® **Dual Duty** **Pulleys**



RECOMMENDED MINIMUM PULLEY PITCH DIAMETERS

V / WEDGE BELT SECTION	A	B	C	SPZ	SPA	SPB	SPC
	80	125	200	56	90	160	224

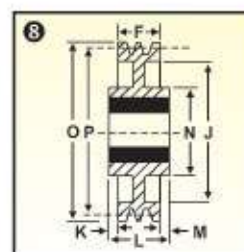
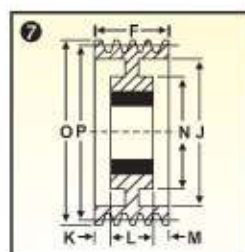
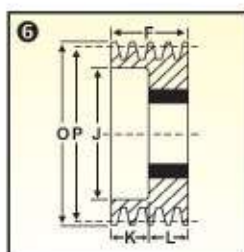
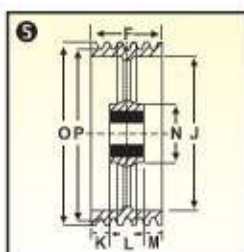
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FOR USE WITH 'SPZ' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
001Z0067	67	1	1108	28	1 $\frac{1}{8}"$	1	16	-	-	22	6	60	72.5
2Z0067		2	1108	28	1 $\frac{1}{8}"$	6	28	38	6.0	22	-	-	
3Z0067		3	1108	28	1 $\frac{1}{8}"$	6	40	38	18.0	22	-	-	
001Z0071	71	1	1108	28	1 $\frac{1}{8}"$	1	16	-	-	22	6.0	60	76.5
2Z0071		2	1108	28	1 $\frac{1}{8}"$	6	28	42	6.0	22	-	-	
3Z0071		3	1108	28	1 $\frac{1}{8}"$	6	40	42	18.0	22	-	-	
001Z0075	75	1	1108	28	1 $\frac{1}{8}"$	1	16	-	-	22	6.0	60	80.5
2Z0075		2	1210	32	1 $\frac{1}{4}"$	6	28	46	3.0	25	-	-	
3Z0075		3	1210	32	1 $\frac{1}{4}"$	6	40	46	15.0	25	-	-	
001Z0080	80	1	1210	32	1 $\frac{1}{4}"$	1	16	-	-	25	9.0	75	85.5
2Z0080		2	1210	32	1 $\frac{1}{4}"$	6	28	51	3.0	25	-	-	
3Z0080		3	1210	32	1 $\frac{1}{4}"$	6	40	51	15.0	25	-	-	
4Z0080		4	1210	32	1 $\frac{1}{4}"$	6	52	51	27.0	25	-	-	
001Z0085	85	1	1210	32	1 $\frac{1}{4}"$	1	16	-	-	25	9.0	83	90.5
2Z0085		2	1610	42	1 $\frac{5}{8}"$	6	28	56	3.0	25	-	-	
3Z0085		3	1610	42	1 $\frac{5}{8}"$	6	40	56	15.0	25	-	-	
4Z0085		4	1610	42	1 $\frac{5}{8}"$	6	52	56	27.0	25	-	-	
5Z0085		5	1610	42	1 $\frac{5}{8}"$	6	64	56	39.0	25	-	-	
001Z0090	90	1	1210	32	1 $\frac{1}{4}"$	1	16	-	-	25	9.0	83	95.5
2Z0090		2	1610	42	1 $\frac{5}{8}"$	6	28	61	3.0	25	-	-	
3Z0090		3	1610	42	1 $\frac{5}{8}"$	6	40	61	15.0	25	-	-	
4Z0090		4	1610	42	1 $\frac{5}{8}"$	6	52	61	27.0	25	-	-	
5Z0090		5	1610	42	1 $\frac{5}{8}"$	6	64	61	39.0	25	-	-	
6Z0090		6	1610	42	1 $\frac{5}{8}"$	6	76	61	51.0	25	-	-	
001Z0095	95	1	1210	32	1 $\frac{1}{4}"$	1	16	-	-	25	9.0	83	100.5
2Z0095		2	1610	42	1 $\frac{5}{8}"$	6	28	66	3.0	25	-	-	
3Z0095		3	1610	42	1 $\frac{5}{8}"$	6	40	66	15.0	25	-	-	
4Z0095		4	1610	42	1 $\frac{5}{8}"$	6	52	66	27.0	25	-	-	
5Z0095		5	1610	42	1 $\frac{5}{8}"$	6	64	66	39.0	25	-	-	
6Z0095		6	1610	42	1 $\frac{5}{8}"$	6	76	66	51.0	25	-	-	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**



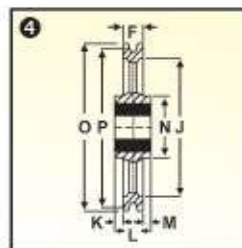
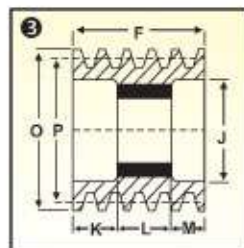
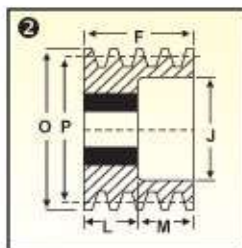
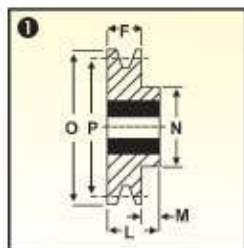
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FOR USE WITH 'SPZ' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
001Z0100	100	1	1210	32	1 $\frac{1}{4}"$	1	16	-	-	25	9	83	105.5
2Z0100		2	1610	42	1 $\frac{5}{8}"$	6	28	71	3.0	25	-	-	
3Z0100		3	1610	42	1 $\frac{5}{8}"$	6	40	71	15.0	25	-	-	
4Z0100		4	2012	50	2"	6	52	71	20.0	32	-	-	
5Z0100		5	2012	50	2"	6	64	71	32.0	32	-	-	
6Z0100		6	2012	50	2"	6	76	71	44.0	32	-	-	
001Z0112	112	1	1610	42	1 $\frac{5}{8}"$	1	16	-	-	25	9.0	92	117.5
2Z0112		2	1610	42	1 $\frac{5}{8}"$	6	28	83	3.0	25	-	-	
3Z0112		3	2012	50	2"	6	40	83	8.0	32	-	-	
4Z0112		4	2012	50	2"	6	52	83	20.0	32	-	-	
5Z0112		5	2012	50	2"	6	64	83	32.0	32	-	-	
6Z0112		6	2012	50	2"	6	76	83	44.0	32	-	-	
001Z0125	125	1	1610	42	1 $\frac{5}{8}"$	1	16	-	-	25	9.0	92	130.5
2Z0125		2	1610	42	1 $\frac{5}{8}"$	6	28	96	3.0	25	-	-	
3Z0125		3	2012	50	2"	6	40	96	8.0	32	-	-	
4Z0125		4	2012	50	2"	6	52	96	20.0	32	-	-	
5Z0125		5	2517	60	2 $\frac{1}{2}"$	6	64	96	19.0	45	-	-	
6Z0125		6	2517	60	2 $\frac{1}{2}"$	6	76	96	31.0	45	-	-	
001Z0140	140	1	1610	42	1 $\frac{5}{8}"$	1	16	-	-	25	9.0	92	145.5
2Z0140		2	1610	42	1 $\frac{5}{8}"$	6	28	111	3	25	-	-	
3Z0140		3	2012	50	2"	2	40	111	-	32	8.0	-	
4Z0140		4	2012	50	2"	2	52	111	-	32	20.0	-	
5Z0140		5	2517	60	2 $\frac{1}{2}"$	2	64	111	-	45	19.0	-	
6Z0140		6	2517	60	2 $\frac{1}{2}"$	2	76	111	-	45	31.0	-	
001Z0160	160	1	1610	42	1 $\frac{5}{8}"$	1	16	-	-	25	9.0	92	165.5
2Z0160		2	2012	50	2"	1	28	-	-	32	4.0	112	
3Z0160		3	2012	50	2"	2	40	131	-	32	8.0	-	
4Z0160		4	2517	60	2 $\frac{1}{2}"$	2	52	131	-	45	7.0	-	
5Z0160		5	2517	60	2 $\frac{1}{2}"$	2	64	131	-	45	19.0	-	
6Z0160		6	2517	60	2 $\frac{1}{2}"$	2	76	131	-	45	31.0	-	
001Z0180	180	1	1610	42	1 $\frac{5}{8}"$	1	16	-	-	25	9.0	92	185.5
2Z0180		2	2012	50	2"	1	28	-	-	32	4.0	112	
3Z0180		3	2012	50	2"	2	40	151	-	32	8.0	-	
4Z0180		4	2517	60	2 $\frac{1}{2}"$	2	52	151	-	45	7.0	-	
5Z0180		5	2517	60	2 $\frac{1}{2}"$	2	64	151	-	45	19.0	-	
6Z0180		6	2517	60	2 $\frac{1}{2}"$	2	76	151	-	45	31.0	-	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**

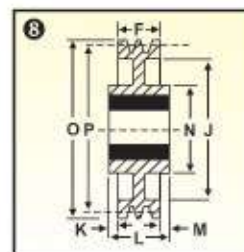
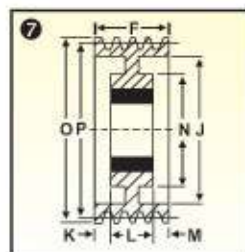
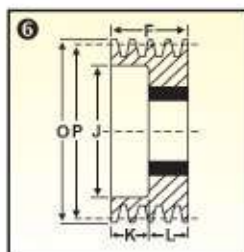
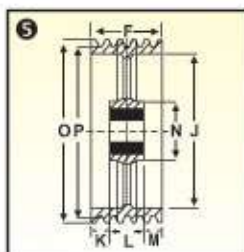


FOR USE WITH 'SPZ' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
001Z0200	200	1	2012	50	2"	1	16	-	-	32.0	16.0	112	205.5
2Z0200		2	2012	50	2"	1	28	171	-	32.0	4.0	124	
3Z0200		3	2012	50	2"	2	40	171	-	32.0	8.0	124	
4Z0200		4	2517	60	2 1/2"	3	52	171	3.5	45.0	3.5	-	
5Z0200		5	2517	60	2 1/2"	7	64	171	9.5	45.0	9.5	124	
6Z0200		6	2517	60	2 1/2"	7	76	171	15.5	45.0	15.5	124	
001Z0250	250	1	2012	50	2"	4	16	221	8.0	32.0	8.0	112	255.5
2Z0250		2	2012	50	2"	4	28	221	2.0	32.0	2.0	112	
3Z0250		3	2012	50	2"	5	40	221	4.0	32.0	4.0	112	
4Z0250		4	2517	60	2 1/2"	7	52	221	3.5	45.0	3.5	124	
5Z0250		5	2517	60	2 1/2"	5	64	221	9.5	45.0	9.5	124	
6Z0250		6	2517	60	2 1/2"	5	76	221	15.5	45.0	15.5	124	
001Z0315	315	1	2012	50	2"	4	16	286	8.0	32.0	8.0	112	320.5
2Z0315		2	2012	50	2"	4	28	286	2.0	32.0	2.0	112	
3Z0315		3	2517	60	2 1/2"	8	40	286	2.5	45.0	2.5	124	
4Z0315		4	2517	60	2 1/2"	5	52	286	3.5	45.0	3.5	124	
5Z0315		5	2517	60	2 1/2"	5	64	286	9.5	45.0	9.5	124	
6Z0315		6	2517	60	2 1/2"	5	76	286	15.5	45.0	15.5	124	
001Z0400	400	1	2012	50	2"	4	16	371	8.0	32.0	8.0	112	405.5
2Z0400		2	2517	60	2 1/2"	4	28	371	8.5	45.0	8.5	124	
3Z0400		3	2517	60	2 1/2"	4	40	371	2.5	45.0	2.5	124	
4Z0400		4	2517	60	2 1/2"	5	52	371	3.5	45.0	3.5	124	
5Z0400		5	3020	75	3"	5	64	371	6.5	51.0	6.5	159	
6Z0400		6	3030	75	3"	4	76	371	-	76.0	-	159	
002Z0500	500	2	2517	60	2 1/2"	4	28	471	8.5	45.0	8.5	124	505.5
3Z0500		3	2517	60	2 1/2"	4	40	471	2.5	45.0	2.5	124	
4Z0500		4	3020	75	3"	5	52	471	0.5	51.0	0.5	159	
5Z0500		5	3020	75	3"	5	64	471	6.5	51.0	6.5	159	
6Z0500		6	3030	75	3"	4	76	471	-	76.0	-	159	
003Z0630	630	3	2517	60	2 1/2"	4	40	601	2.5	45.0	2.5	124	635.5
4Z0630		4	3020	75	3"	5	52	601	0.5	51.0	0.5	159	
5Z0630		5	3030	75	3"	4	64	601	6.0	76.0	6.0	159	
6Z0630		6	3535	90	3 1/2"	4	76	601	6.5	89.0	6.5	178	
003Z0800	800	3	3030	75	3"	4	40	771	18.0	76.0	18.0	159	805.5
4Z0800		4	3030	75	3"	4	52	771	12.0	76.0	12.0	159	
5Z0800		5	3535	90	3 1/2"	4	64	771	12.5	89.0	12.5	178	
6Z0800		6	3535	90	3 1/2"	4	76	771	6.5	89.0	6.5	178	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

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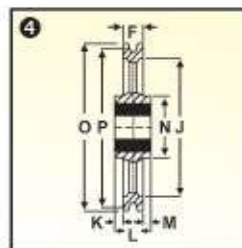
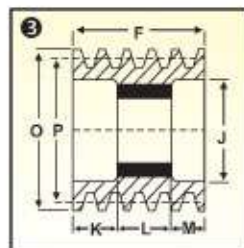
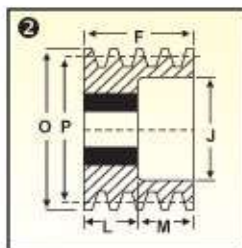
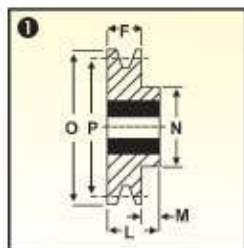


FOR USE WITH 'A' & 'SPA' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
FOR USE WITH 'A' SECTION BELTS													
002A0080	80	2	1108	28	1 ¼"	6	35	50	13	22	-	-	85.5
3A0080		3	1210	32	1 ¼"	6	50	50	25	25	-	-	
4A0080		4	1215	32	1 ¼"	6	65	50	27	38	-	-	
5A0080		5	1215	32	1 ¼"	6	80	50	42	38	-	-	
002A0085	85	2	1108	28	1 ¼"	6	35	55	13	22	-	-	90.5
3A0085		3	1210	32	1 ¼"	6	50	55	25	25	-	-	
4A0085		4	1215	32	1 ¼"	6	65	55	27	38	-	-	
5A0085		5	1215	32	1 ¼"	6	80	55	42	38	-	-	
002A0090	90	2	1108	28	1 ¼"	6	35	60	13.0	22	-	-	95.5
3A0090		3	1610	42	1 ¾"	6	50	60	25.0	25	-	-	
4A0090		4	1615	42	1 ¾"	3	65	60	13.5	38	13.5	-	
5A0090		5	1615	42	1 ¾"	3	80	60	21.0	38	21.0	-	
FOR USE WITH 'A' / 'SPA' SECTION BELTS													
002A0095	95	2	1610	42	1 ¾"	6	35	65	10.0	25	-	-	100.5
3A0095		3	1610	42	1 ¾"	6	50	65	25.0	25	-	-	
4A0095		4	1615	42	1 ¾"	3	65	65	13.5	38	13.5	-	
5A0095		5	1615	42	1 ¾"	3	80	65	21.0	38	21.0	-	
002A0100	100	2	1610	42	1 ¾"	6	35	65	10.0	25	-	-	105.5
3A0100		3	1610	42	1 ¾"	6	50	65	25.0	25	-	-	
4A0100		4	1615	42	1 ¾"	6	65	65	27.0	38	-	-	
5A0100		5	1615	42	1 ¾"	6	80	65	42.0	38	-	-	
002A0106	106	2	1610	42	1 ¾"	6	35	71	10.0	25	-	-	111.5
3A0106		3	1610	42	1 ¾"	6	50	71	25.0	25	-	-	
4A0106		4	1615	42	1 ¾"	6	65	71	27.0	38	-	-	
5A0106		5	2012	50	2"	6	80	71	48.0	32	-	-	
002A0112	112	2	1610	42	1 ¾"	6	35	77	10.0	25	-	-	117.5
3A0112		3	2012	50	2"	6	50	77	18.0	32	-	-	
4A0112		4	2012	50	2"	6	65	77	33.0	32	-	-	
5A0112		5	2012	50	2"	6	80	77	48.0	32	-	-	
002A0118	118	2	1610	42	1 ¾"	6	35	84	10.0	25	-	-	123.5
3A0118		3	2012	50	2"	6	50	84	18.0	32	-	-	
4A0118		4	2012	50	2"	6	65	84	33.0	32	-	-	
5A0118		5	2012	50	2"	6	80	84	48.0	32	-	-	
002A0125	125	2	1610	42	1 ¾"	2	35	90	-	25	10.0	-	130.5
3A0125		3	2012	50	2"	2	50	90	-	32	18.0	-	
4A0125		4	2012	50	2"	2	65	90	-	32	33.0	-	
5A0125		5	2012	50	2"	2	80	90	-	32	48.0	-	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**

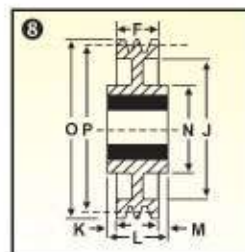
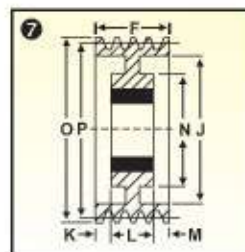
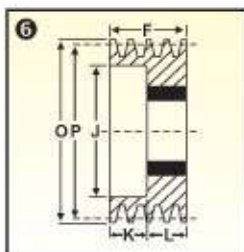
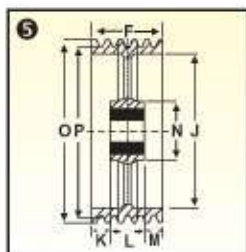


FOR USE WITH 'A' / 'SPA' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
002A0132	132	2	1610	42	1 3/4"	2	35	98	-	25	10.0	-	137.5
3A0132		3	2012	50	2"	2	50	98	-	32	18.0	-	
4A0132		4	2517	60	2 1/2"	2	65	98	-	45	20.0	-	
5A0132		5	2517	60	2 1/2"	2	80	98	-	45	35.0	-	
002A0140	140	2	2012	50	2"	6	35	105	3	32	-	-	145.5
3A0140		3	2517	60	2 1/2"	2	50	105	-	45	5.0	-	
4A0140		4	2517	60	2 1/2"	2	65	105	-	45	20.0	-	
5A0140		5	2517	60	2 1/2"	2	80	105	-	45	35.0	-	
002A0150	150	2	2012	50	2"	6	35	115	3.0	32	-	-	155.5
3A0150		3	2517	60	2 1/2"	2	50	115	-	45	5.0	-	
4A0150		4	2517	60	2 1/2"	2	65	115	-	45	20.0	-	
5A0150		5	2517	60	2 1/2"	2	80	115	-	45	35.0	-	
002A0160	160	2	2012	50	2"	6	35	125	3.0	32	-	-	165.5
3A0160		3	2517	60	2 1/2"	2	50	125	-	45	5.0	-	
4A0160		4	2517	60	2 1/2"	2	65	125	-	45	20.0	-	
5A0160		5	2517	60	2 1/2"	2	80	125	-	45	35.0	-	
002A0180	180	2	2012	50	2"	2	35	145	-	32	3.0	-	185.5
3A0180		3	2517	60	2 1/2"	2	50	145	-	45	5.0	-	
4A0180		4	2517	60	2 1/2"	2	65	145	-	45	20.0	-	
5A0180		5	3020	75	3"	2	80	145	-	51	29.0	-	
002A0200	200	2	2517	60	2 1/2"	8	35	165	-	45	10.0	124	205.5
3A0200		3	2517	60	2 1/2"	3	50	165	2.5	45	2.5	-	
4A0200		4	3020	75	3"	2	65	165	-	51	14.0	-	
5A0200		5	3020	75	3"	2	80	165	-	51	29.0	-	
002A0250	250	2	2517	60	2 1/2"	8	35	215	-	45	10.0	124	255.5
3A0250		3	2517	60	2 1/2"	7	50	215	2.5	45	2.5	124	
4A0250		4	3020	75	3"	7	65	215	7.0	51	7.0	159	
5A0250		5	3020	75	3"	7	80	215	14.5	51	14.5	159	
002A0315	315	2	2517	60	2 1/2"	4	35	280	-	45	10.0	124	320.5
3A0315		3	3020	75	3"	8	50	280	0.5	51	0.5	159	
4A0315		4	3020	75	3"	7	65	280	7.0	51	7.0	159	
5A0315		5	3535	90	3 1/2"	8	80	280	4.5	89	4.5	178	
002A0400	400	2	2517	60	2 1/2"	4	35	365	-	45	10.0	124	405.5
3A0400		3	3020	75	3"	4	50	365	0.5	51	0.5	159	
4A0400		4	3020	75	3"	5	65	365	7.0	51	7.0	159	
5A0400		5	3535	90	3 1/2"	4	80	365	4.5	89	4.5	178	
002A0500	500	2	2517	60	2 1/2"	4	35	465	-	45	10.0	124	505.5
3A0500		3	3020	75	3"	4	50	465	0.5	51	0.5	159	
4A0500		4	3020	75	3"	5	65	465	7.0	51	7.0	159	
5A0500		5	3535	90	3 1/2"	4	80	465	4.5	89	4.5	178	
002A0630	630	2	3020	75	3"	4	35	595	-	51	16.0	159	635.5
3A0630		3	3020	75	3"	4	50	595	0.5	51	0.5	159	
4A0630		4	3535	90	3 1/2"	4	65	595	12.0	89	12.0	178	
5A0630		5	3535	90	3 1/2"	4	80	595	4.5	89	4.5	178	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**

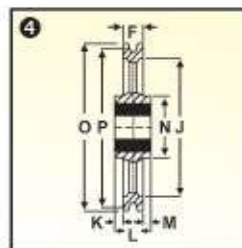
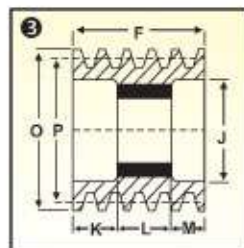
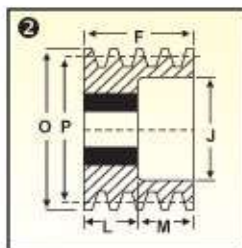
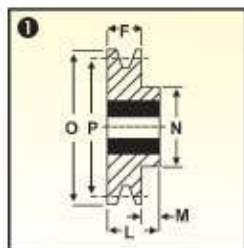


FOR USE WITH 'B' & 'SPB' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
FOR USE WITH 'B' SECTION BELTS													
002B0125	125	2	1610	42	1 ¾"	6	44	85	19.0	25	-	-	132
3B0125		3	1610	42	1 ¾"	6	63	85	38.0	25	-	-	
4B0125		4	2012	50	2"	3	82	89	25.0	32	25.0	-	
5B0125		5	2012	50	2"	3	101	89	34.5	32	34.5	-	
6B0125		6	2012	50	2"	3	120	89	44.0	32	44.0	-	
002B0132	132	2	1610	42	1 ¾"	6	44	92	19.0	25	-	-	139
3B0132		3	1610	42	1 ¾"	6	63	92	38.0	25	-	-	
4B0132		4	2012	50	2"	3	82	92	25.0	32	25.0	-	
5B0132		5	2012	50	2"	2	101	92	-	32	69.0	-	
6B0132		6	2012	50	2"	3	120	92	44.0	32	44.0	-	
002B0140	140	2	1610	42	1 ¾"	6	44	100	19.0	25	-	-	147
3B0140		3	1610	42	1 ¾"	6	63	100	38.0	25	-	-	
4B0140		4	2012	50	2"	3	82	100	25.0	32	25.0	-	
5B0140		5	2012	50	2"	2	101	100	-	32	69.0	-	
6B0140		6	2517	60	2 ½"	3	120	109	37.5	45	37.5	-	
002B0150	150	2	1610	42	1 ¾"	6	44	110	19.0	25	-	-	157
3B0150		3	2012	50	2"	6	63	110	31.0	32	-	-	
4B0150		4	2012	50	2"	3	82	110	25.0	32	25.0	-	
5B0150		5	2517	60	2 ½"	2	101	110	-	45	56.0	-	
6B0150		6	2517	60	2 ½"	2	120	110	-	45	75.0	-	
FOR USE WITH 'B' / 'SPB' SECTION BELTS													
002B0160	160	2	2012	50	2"	6	44	114	12.0	32	-	-	167
3B0160		3	2517	60	2 ½"	6	63	114	18.0	45	-	-	
4B0160		4	2517	60	2 ½"	3	82	114	18.5	45	18.5	-	
5B0160		5	2517	60	2 ½"	3	101	114	28.0	45	28.0	-	
6B0160		6	2517	60	2 ½"	2	120	114	-	45	75.0	-	
002B0170	170	2	2012	50	2"	6	44	124	12.0	32	-	-	177
3B0170		3	2517	60	2 ½"	6	63	124	18.0	45	-	-	
4B0170		4	2517	60	2 ½"	3	82	124	18.5	45	18.5	-	
5B0170		5	3020	75	3"	3	101	128	25.0	51	25.0	-	
6B0170		6	3020	75	3"	2	120	128	-	51	69.0	-	
002B0180	180	2	2012	50	2"	6	44	134	12.0	32	-	-	187
3B0180		3	2517	60	2 ½"	6	63	134	18.0	45	-	-	
4B0180		4	2517	60	2 ½"	3	82	134	18.5	45	18.5	-	
5B0180		5	3020	75	3"	3	101	134	25.0	51	25.0	-	
6B0180		6	3020	75	3"	2	120	134	-	51	69.0	-	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**

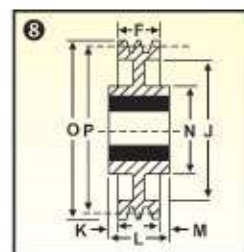
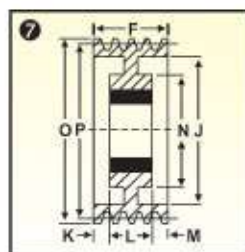
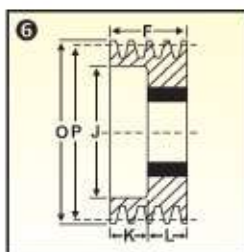
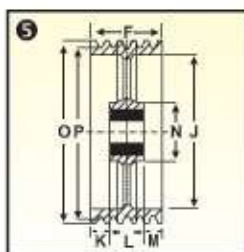


FOR USE WITH 'B' / 'SPB' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
002B0190	190	2	2517	60	2 ½"	1	44	-	-	44	-	-	197
3B0190		3	2517	60	2 ½"	6	63	144	18.0	45	-	-	
4B0190		4	2517	60	2 ½"	3	82	144	18.5	45	18.5	-	
5B0190		5	3020	75	3"	3	101	144	25.0	51	25.0	-	
6B0190		6	3020	75	3"	2	120	144	-	51	69.0	-	
002B0200	200	2	2517	60	2 ½"	1	44	-	-	44	-	-	207
3B0200		3	2517	60	2 ½"	2	63	154	-	45	18.0	-	
4B0200		4	3020	75	3"	2	82	154	-	51	31.0	-	
5B0200		5	3020	75	3"	2	101	154	-	51	50.0	-	
6B0200		6	3020	75	3"	2	120	154	-	51	69.0	-	
002B0212	212	2	2517	60	2 ½"	1	44	-	-	44	-	-	219
3B0212		3	2517	60	2 ½"	2	63	166	-	45	18.0	-	
4B0212		4	3020	75	3"	2	82	166	-	51	31.0	-	
5B0212		5	3020	75	3"	2	101	166	-	51	50.0	-	
6B0212		6	3020	75	3"	2	120	166	-	51	69.0	-	
002B0224	224	2	2517	60	2 ½"	1	44	-	-	44	-	-	231
3B0224		3	2517	60	2 ½"	2	63	178	-	45	18.0	-	
4B0224		4	3020	75	3"	2	82	178	-	51	31.0	-	
5B0224		5	3020	75	3"	2	101	178	-	51	50.0	-	
6B0224		6	3020	75	3"	2	120	178	-	51	69.0	-	
002B0236	236	2	2517	60	2 ½"	1	44	-	-	44	-	-	243
3B0236		3	2517	60	2 ½"	2	63	190	-	45	18.0	-	
4B0236		4	3020	75	3"	2	82	190	-	51	31.0	-	
5B0236		5	3020	75	3"	2	101	190	-	51	50.0	-	
6B0236		6	3020	75	3"	2	120	190	-	51	69.0	-	
002B0250	250	2	2517	60	2 ½"	8	44	-	-	44	-	-	257
3B0250		3	3020	75	3"	2	63	204	-	51	12.0	-	
4B0250		4	3020	75	3"	2	82	204	-	51	31.0	-	
5B0250		5	3020	75	3"	2	101	204	-	51	50.0	-	
6B0250		6	3020	75	3"	2	120	204	-	51	69.0	-	
002B0280	280	2	2517	60	2 ½"	8	44	234	0.5	45	0.5	124	287
3B0280		3	3020	75	3"	7	63	234	6.0	51	6.0	159	
4B0280		4	3020	75	3"	7	82	234	15.5	51	15.5	159	
5B0280		5	3535	90	3 ½"	7	101	234	6.0	89	6.0	178	
6B0280		6	3535	90	3 ½"	7	120	234	15.5	89	15.5	178	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**

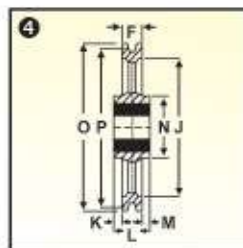
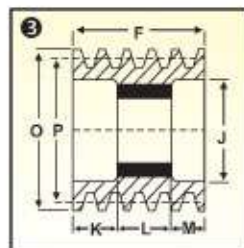
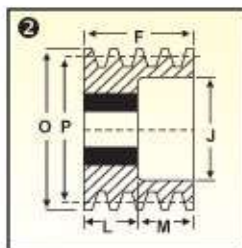
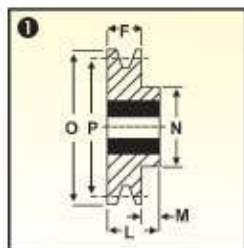


FOR USE WITH 'B' / 'SPB' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
002B0315	315	2	2517	60	2 1/2"	8	44	269	0.5	45	0.5	124	322
3B0315		3	3020	75	3"	7	63	269	6.0	51	6.0	159	
4B0315		4	3535	90	3 1/2"	8	82	269	3.5	89	3.5	178	
5B0315		5	3535	90	3 1/2"	7	101	269	6.0	89	6.0	178	
6B0315		6	3535	90	3 1/2"	7	120	269	15.5	89	15.5	178	
002B0355	355	2	3020	75	3"	4	44	309	3.5	51	3.5	159	362
3B0355		3	3020	75	3"	5	63	309	6.0	51	6.0	159	
4B0355		4	3535	90	3 1/2"	8	82	309	3.5	89	3.5	178	
5B0355		5	3535	90	3 1/2"	7	101	309	6.0	89	6.0	178	
6B0355		6	3535	90	3 1/2"	7	120	309	15.5	89	15.5	178	
002B0400	400	2	3020	75	3"	4	44	354	3.5	51	3.5	159	407
3B0400		3	3535	90	3 1/2"	4	63	354	13.0	89	13.0	178	
4B0400		4	3535	90	3 1/2"	4	82	354	3.5	89	3.5	178	
5B0400		5	3535	90	3 1/2"	5	101	354	6.0	89	6.0	178	
6B0400		6	3535	90	3 1/2"	5	120	354	15.5	89	15.5	178	
002B0500	500	2	3020	75	3"	4	44	454	3.5	51	3.5	159	507
3B0500		3	3535	90	3 1/2"	4	63	454	13.0	89	13.0	178	
4B0500		4	3535	90	3 1/2"	4	82	454	3.5	89	3.5	178	
5B0500		5	3535	90	3 1/2"	5	101	454	6.0	89	6.0	178	
6B0500		6	4040	100	4"	5	120	454	9.0	102	9.0	216	
002B0630	630	2	3030	75	3"	4	44	584	16.0	76	16.0	159	637
3B0630		3	3535	90	3 1/2"	4	63	584	13.0	89	13.0	178	
4B0630		4	3535	90	3 1/2"	4	82	584	3.5	89	3.5	178	
5B0630		5	4040	100	4"	4	101	584	0.5	102	0.5	216	
6B0630		6	4040	100	4"	5	120	584	9.0	102	9.0	216	
002B0800	800	2	3030	75	3"	4	44	754	16.0	76	16.0	159	807
3B0800		3	3535	90	3 1/2"	4	63	754	13.0	89	13.0	178	
4B0800		4	4040	100	4"	4	82	754	10.0	102	10.0	216	
5B0800		5	4040	100	4"	4	101	754	0.5	102	0.5	216	
6B0800		6	4545	110	4 1/2"	5	120	754	3.0	114	3.0	242	
003B1000	1000	3	4040	100	4"	4	63	954	19.5	102	19.5	216	1007
4B1000		4	4040	100	4"	4	82	954	10.0	102	10.0	216	
5B1000		5	4545	110	4 1/2"	4	101	954	6.5	114	6.5	242	
6B1000		6	4545	110	4 1/2"	5	120	954	3.0	114	3.0	242	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**

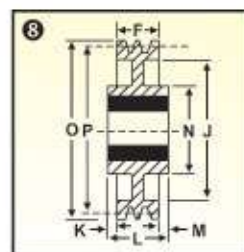
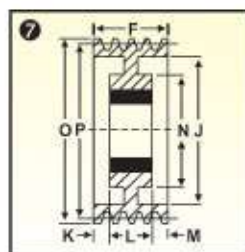
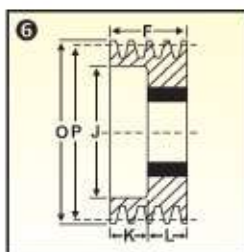
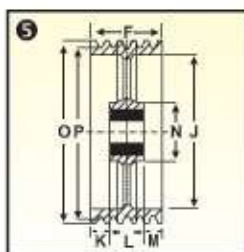


FOR USE WITH 'C' & 'SPC' SECTION BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
FOR USE WITH 'C' SECTION BELTS													
004C0200	200	4	3020	75	3"	3	111	151	30.0	51	30.0	-	209.6
5C0200		5	3535	90	3 ½"	3	136	151	16.0	89	31.0	-	
6C0200		6	3535	90	3 ½"	3	162	151	16.0	89	57.0	-	
7C0200		7	3535	90	3 ½"	3	187	151	36.0	89	62.0	-	
8C0200		8	3535	90	3 ½"	3	213	151	46.0	89	78.0	-	
004C0212	212	4	3020	75	3"	3	111	163	30.0	51	30.0	-	221.6
5C0212		5	3535	90	3 ½"	3	136	163	16.0	89	31.0	-	
6C0212		6	3535	90	3 ½"	3	162	163	36.5	89	36.5	-	
7C0212		7	3535	90	3 ½"	3	187	163	36.0	89	62.0	-	
8C0212		8	3535	90	3 ½"	3	213	163	46.0	89	78.0	-	
FOR USE WITH 'C' / 'SPC' SECTION BELTS													
004C0224	224	4	3535	90	3 ½"	3	111	161	11.0	89	11.0	-	233.6
5C0224		5	3535	90	3 ½"	3	136	161	23.5	89	23.5	-	
6C0224		6	3535	90	3 ½"	3	162	161	36.5	89	36.5	-	
7C0224		7	3535	90	3 ½"	3	187	161	36.0	89	62.0	-	
8C0224		8	3535	90	3 ½"	3	213	161	46.0	89	78.0	-	
004C0236	236	4	3535	90	3 ½"	3	111	173	11.0	89	11.0	-	245.6
5C0236		5	3535	90	3 ½"	3	136	173	23.5	89	23.5	-	
6C0236		6	3535	90	3 ½"	3	162	173	36.5	89	36.5	-	
7C0236		7	3535	90	3 ½"	3	187	173	36.0	89	62.0	-	
8C0236		8	3535	90	3 ½"	3	213	173	46.0	89	78.0	-	
004C0250	250	4	3535	90	3 ½"	3	111	187	11.0	89	11.0	-	259.6
5C0250		5	3535	90	3 ½"	3	136	187	23.5	89	23.5	-	
6C0250		6	3535	90	3 ½"	3	162	187	36.5	89	36.5	-	
7C0250		7	3535	90	3 ½"	3	187	187	36.0	89	62.0	-	
8C0250		8	3535	90	3 ½"	3	213	187	62.0	89	62.0	-	
004C0265	265	4	3535	90	3 ½"	3	111	202	11.0	89	11.0	-	274.6
5C0265		5	3535	90	3 ½"	3	136	202	23.5	89	23.5	-	
6C0265		6	3535	90	3 ½"	3	162	202	36.5	89	36.5	-	
7C0265		7	3535	90	3 ½"	3	187	202	36.0	89	62.0	-	
8C0265		8	3535	90	3 ½"	3	213	202	62.0	89	62.0	-	
004C0280	280	4	3535	90	3 ½"	3	111	217	11.0	89	11.0	-	289.6
5C0280		5	3535	90	3 ½"	3	136	217	23.5	89	23.5	-	
6C0280		6	3535	90	3 ½"	3	162	217	36.5	89	36.5	-	
7C0280		7	3535	90	3 ½"	3	187	217	36.0	89	62.0	-	
8C0280		8	3535	90	3 ½"	3	213	217	62.0	89	62.0	-	
004C0300	300	4	3535	90	3 ½"	7	111	237	11.0	89	11.0	178	309.6
5C0300		5	3535	90	3 ½"	7	136	237	23.5	89	23.5	178	
6C0300		6	3535	90	3 ½"	7	162	237	36.5	89	36.5	178	
7C0300		7	3535	90	3 ½"	7	187	237	49.0	89	49.0	178	
8C0300		8	4040	100	4"	3	213	237	55.5	102	55.5	-	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® Dual Duty Pulleys

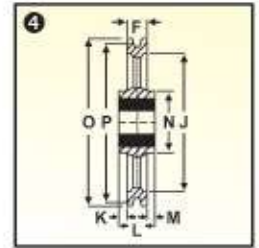
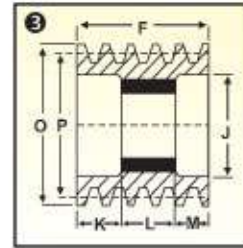
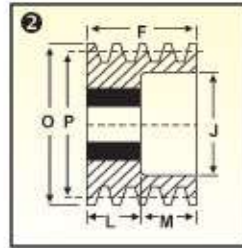
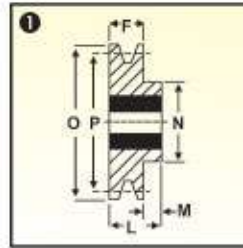


FOR USE WITH 'C' / 'SPC' BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
004C0315	315	4	3535	90	3 1/2"	7	111	252	11.0	89	11.0	178	324.6
5C0315		5	3535	90	3 1/2"	7	136	252	23.5	89	23.5	178	
6C0315		6	3535	90	3 1/2"	7	162	252	36.5	89	36.5	178	
7C0315		7	3535	90	3 1/2"	7	187	252	49.0	89	49.0	178	
8C0315		8	4040	100	4"	3	213	252	55.5	102	55.5	-	
004C0335	335	4	3535	90	3 1/2"	7	111	272	11.0	89	11.0	178	344.6
5C0335		5	3535	90	3 1/2"	7	136	272	23.5	89	23.5	178	
6C0335		6	3535	90	3 1/2"	7	162	272	36.5	89	36.5	178	
7C0335		7	3535	90	3 1/2"	7	187	272	49.0	89	49.0	178	
8C0335		8	4040	100	4"	3	213	272	55.5	102	55.5	-	
004C0355	355	4	3535	90	3 1/2"	7	111	292	11.0	89	11.0	178	364.6
5C0355		5	3535	90	3 1/2"	7	136	292	23.5	89	23.5	178	
6C0355		6	3535	90	3 1/2"	7	162	292	36.5	89	36.5	178	
7C0355		7	4040	100	4"	7	187	292	42.5	102	42.5	216	
8C0355		8	4040	100	4"	7	213	292	55.5	102	55.5	216	
004C0375	375	4	3535	90	3 1/2"	7	111	312	11.0	89	11.0	178	384.6
5C0375		5	3535	90	3 1/2"	7	136	312	23.5	89	23.5	178	
6C0375		6	4040	100	4"	7	162	312	30.0	102	30.0	216	
7C0375		7	4040	100	4"	7	187	312	42.5	102	42.5	216	
8C0375		8	4545	110	4 1/2"	7	213	312	49.5	114	49.5	242	
004C0400	400	4	3535	90	3 1/2"	5	111	337	11.0	89	11.0	178	409.6
5C0400		5	3535	90	3 1/2"	5	136	337	23.5	89	23.5	178	
6C0400		6	4040	100	4"	7	162	337	30.0	102	30.0	216	
7C0400		7	4545	110	4 1/2"	7	187	337	36.5	114	36.5	242	
8C0400		8	4545	110	4 1/2"	7	213	337	49.5	114	49.5	242	
004C0425	425	4	3535	90	3 1/2"	5	111	362	11.0	89	11.0	178	434.6
5C0425		5	4040	100	4"	7	136	362	17.0	102	17.0	216	
6C0425		6	4545	110	4 1/2"	7	162	362	24.0	114	24.0	242	
7C0425		7	4545	110	4 1/2"	7	187	362	36.5	114	36.5	242	
8C0425		8	5050	125	5"	7	213	362	43.0	127	43.0	267	
004C0450	450	4	3535	90	3 1/2"	5	111	387	11.0	89	11.0	178	459.6
5C0450		5	4040	100	4"	5	136	387	17.0	102	17.0	216	
6C0450		6	4545	110	4 1/2"	7	162	387	24.0	114	24.0	242	
7C0450		7	5050	125	5"	7	187	387	30.0	127	30.0	267	
8C0450		8	5050	125	5"	7	213	387	43.0	127	43.0	267	
004C0475	475	4	3535	90	3 1/2"	5	111	412	11.0	89	11.0	178	484.6
5C0475		5	4040	100	4"	5	136	412	17.0	102	17.0	216	
6C0475		6	4545	110	4 1/2"	7	162	412	24.0	114	24.0	242	
7C0475		7	5050	125	5"	7	187	412	30.0	127	30.0	267	
8C0475		8	5050	125	5"	7	213	412	43.0	127	43.0	267	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Taper Lock® **Dual Duty** **Pulleys**



FOR USE WITH 'C' / 'SPC' BELTS

Product Code	Pitch Dia(P)	No. of Grooves	Bush No.	Max. Bore		Pulley Type	F	J	K	L	M	N	Outside Dia (O)
				Met.	Imp.								
004C0500	500	4	3535	90	3 1/2"	5	111	437	11.0	89	11.0	178	509.6
5C0500		5	4040	100	4"	5	136	437	17.0	102	17.0	216	
6C0500		6	4545	110	4 1/2"	5	162	437	24.0	114	24.0	242	
7C0500		7	5050	125	5"	7	187	437	30.0	127	30.0	267	
8C0500		8	5050	125	5"	7	213	437	43.0	127	43.0	267	
004C0530	530	4	4040	100	4"	5	111	467	4.5	102	4.5	216	539.6
5C0530		5	4545	110	4 1/2"	5	136	467	11.0	114	11.0	242	
6C0530		6	5050	125	5"	5	162	467	17.5	127	17.5	267	
7C0530		7	5050	125	5"	7	187	467	30.0	127	30.0	267	
8C0530		8	5050	125	5"	7	213	467	43.0	127	43.0	267	
004C0560	560	4	4040	100	4"	5	111	497	4.5	102	4.5	216	569.6
5C0560		5	4545	110	4 1/2"	5	136	497	11.0	114	11.0	242	
6C0560		6	5050	125	5"	5	162	497	17.5	127	17.5	267	
7C0560		7	5050	125	5"	5	187	497	30.0	127	30.0	267	
8C0560		8	5050	125	5"	5	213	497	43.0	127	43.0	267	
004C0630	630	4	4545	110	4 1/2"	4	111	567	1.5	114	1.5	242	639.6
5C0630		5	5050	125	5"	5	136	567	4.5	127	4.5	267	
6C0630		6	5050	125	5"	5	162	567	17.5	127	17.5	267	
7C0630		7	5050	125	5"	5	187	567	30.0	127	30.0	267	
8C0630		8	5050	125	5"	5	213	567	43.0	127	43.0	267	
004C0800	800	4	5050	125	5"	4	111	737	8.0	127	8.0	267	809.6
5C0800		5	5050	125	5"	5	136	737	4.5	127	4.5	267	
6C0800		6	5050	125	5"	5	162	737	17.5	127	17.5	267	
7C0800		7	5050	125	5"	5	187	737	30.0	127	30.0	267	
8C0800		8	5050	125	5"	5	213	737	43.0	127	43.0	267	
004C1000	1000	4	5050	125	5"	4	111	937	8.0	127	8.0	267	1009.6
5C1000		5	5050	125	5"	5	136	937	4.5	127	4.5	267	
6C1000		6	5050	125	5"	5	162	937	17.5	127	17.5	267	
7C1000		7	5050	125	5"	5	187	937	30.0	127	30.0	267	
8C1000		8	5050	125	5"	5	213	937	43.0	127	43.0	267	
004C1250	1250	4	5050	125	5"	4	111	1187	8.0	127	8.0	267	1259.6
5C1250		5	5050	125	5"	5	136	1187	4.5	127	4.5	267	
6C1250		6	5050	125	5"	5	162	1187	17.5	127	17.5	267	
7C1250		7	5050	125	5"	5	187	1187	30.0	127	30.0	267	
8C1250		8	5050	125	5"	5	213	1187	43.0	127	43.0	267	

DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED

Fenner

FENAFLEX TYRE COUPLINGS



POWERTRAN

Dependable Power Transmission Accessories

FENNER TYRE COUPLINGS



Fenner Tyre couplings provide all the desirable features of an ideal flexible coupling, including Taper-Lock® fixing. The Fenner Tyre coupling is a "torsionally elastic" coupling offering versatility to designers and engineers with a choice of flange combinations to suit most applications.

The flanges are available in either F or H version Taper-Lock® fitting or bored to size B.

With the addition of a spacer flange, the coupling can be used to accommodate standard distance between shaft ends and facilitate pump maintenance.

Fenner Tyre couplings can accommodate simultaneous maximum misalignment in all planes without imposing undue loads on adjacent bearings and the excellent shock-absorbing properties of the flexible tyre reduce vibrations and torsional oscillations.

Fenner tyres are available in natural rubber compounds for use in ambient temperatures between -50°C to +50°C. Neoprene rubber compounds are available for use in adverse operating conditions e.g. oil or grease contaminations and can be used in temperatures of -15°C to +70°C.

F.R.A.S. tyres are available for use when fire-resistance and antistatic (F.R.A.S.) properties are required.

TORQUE-BORE RANGE

The range includes couplings with torque capacity upto 12606 Nm. and bore diameters upto 190 mm.

MISALIGNMENT

Handles parallel, angular and axial displacements, either singly or in any combination. They can accommodate parallel misalignment upto 6 mm, angular misalignment upto 4° and end float upto 8 mm.

TORSIONALLY SOFT

Cushions against destructive shock loads protecting the complete system, preventing expensive breakdowns and lengthens machine life.

FREE OF BACKLASH

Does not create 'snatch' on take up of the drive.

INSTALLATION

Requires neither special tools nor skilled labour to assemble. Alignment is quickly checked by placing a straight edge across outside diameter of flanges. The split flexible tyre is then positioned in the flanges and the screws tightened into place.

DAMPING

Reduces vibration and torsional oscillations developed in internal combustion engines, the amplitude of which increases greatly at critical points in the speed range. Fenner Tyre coupling dampens these destructive vibrations.

MAINTENANCE

Because there are no moving parts, no lubrication is required. Periodic visual inspection of the tyre is all that is necessary.

EASY ELEMENT REPLACEMENT

To replace flexible element simply loosen the clamping screws, remove the tyre and replace with a new one. It is not necessary to move either driver or driven machine or coupling flanges.

ENVIRONMENT

Use of natural or Neoprene rubber compounds makes the Fenner Tyre coupling suitable for use in most conditions. For fire hazard areas the F.R.A.S. tyre is recommended. This allows free flow of electricity between the two shafts to avoid static electricity build up.

FENNER TYRE COUPLINGS - SELECTION



DETAILS REQUIRED FOR COUPLING SELECTION

1. Type of driven machine and operating hours per day.
2. Speed and power absorbed by driven machine (if absorbed power is not known, calculate on power rating of prime mover).
3. Diameters of shafts to be connected.

PROCEDURE

- a. Service Factor :** Determine the required service factor from table 1.
- b. Design Power :** Multiply the normal running power by the service factor. This gives the design power which is used as a basis for selecting the coupling.
- c. Coupling Size :** Refer to table 3 (page 5) and from the appropriate speed, read across until a power greater than that required in step (b) is found.
The size of Fenner Tyre Coupling required is given at the head of that column.
- d. Bore Size :** Check from dimension tables that chosen flanges can accommodate required bores.

EXAMPLE

A Fenner Tyre Coupling is required to transmit 45 kW from an A.C. Electric Motor which runs at 1440 rev/min to a rotary screen for 12 hours a day. The motor shaft is 60 mm diameter and the screen shaft is 55 mm diameter.

- a. Service Factor :** From table 1, the service factor is 1.4.
- b. Design Power :** Design Power = $45 \times 1.4 = 63$ kW.
- c. Coupling Size :** By reading across from 1440 rev/min in table 3 (page 6) the first power figure to exceed the required 63 kW in step (b) is 76.1 kW. The size of coupling is F100.
- d. Bore Size :** By referring to table 2 (page 5) it can be seen that both shaft diameters fall within the bore range available.

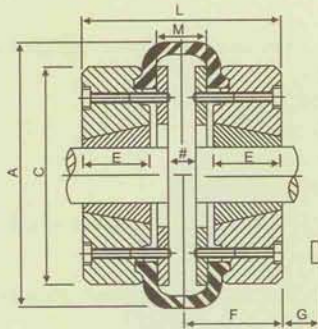
TABLE 1: SERVICE FACTORS

Special Cases	Type of Driving Unit					
For applications where substantial shock, vibration and torque fluctuations occur and for reciprocating machines e.g. internal combustion engines, piston pumps and compressors, refer to Fenner with all machine details for torsion analysis	Electric Motors Steam Turbines			Internal Combustion Engines Steam Engines Water Turbines		
Type of Driven Machine	Operational hours per day					
	10 and under	Over 10 to 16 inclusive	over 16	10 and under	Over 10 to 16 inclusive	over 16
CLASS 1 Agitators, Brewing machinery,Centrifugal compressors and pumps, Belt conveyors, Dynamometers, Line shafts, Fans upto 7.5 kW, Blowers and Exhausters (except positive displacement),Generators.	0.8	0.9	1.0	1.3	1.4	1.5
CLASS 2 Clay working machinery, General Machine tools, Paper mill beaters and winders, Rotary pumps, Rubber extruders, Rotary screens, Textile machinery, Marine propellers and fans over 7.5 kW.	1.3	1.4	1.5	1.8	1.9	2.0
CLASS 3 Bucket elevators, Cooling tower fans, Piston compressors and pumps, Foundry machinery, Metal presses, Paper mill calendars, Pulverisers and Positive displacement blowers.	1.8	1.9	2.0	2.3	2.4	2.5
CLASS 4 Reciprocating conveyors, Gyratory crushers, Mills (ball, pebble and rod), Rubber machinery (Banbury mixers and mills) and Vibratory screens	2.3	2.4	2.5	2.8	2.9	3.0

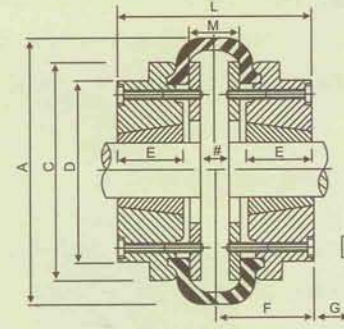
FENNER TYRE COUPLINGS



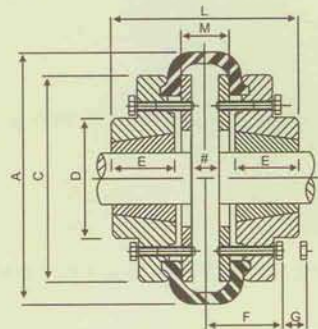
F TYPE



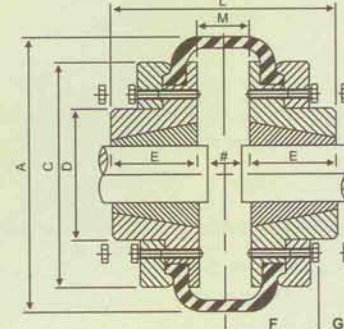
F - 40 & 45



F - 50 & 60

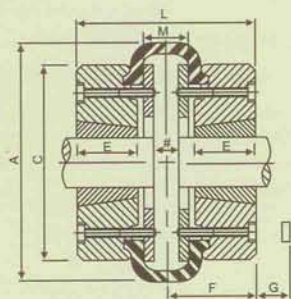


F - 70 TO 120

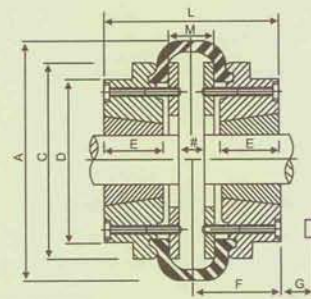


F - 140 TO 220

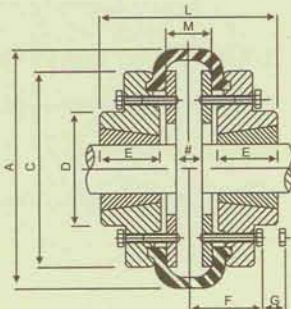
H TYPE



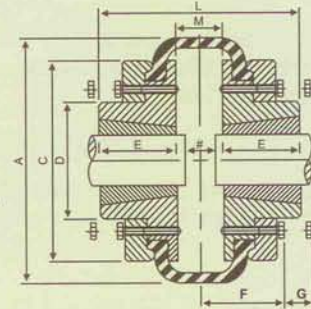
F - 40 & 45



F - 50 & 60



F - 70 TO 120

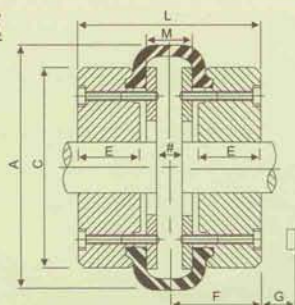


F - 140 TO 220

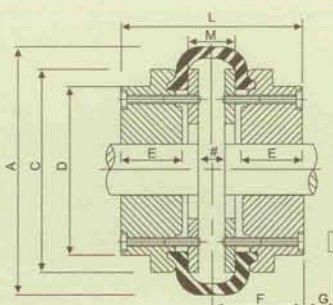
FENNER TYRE COUPLINGS



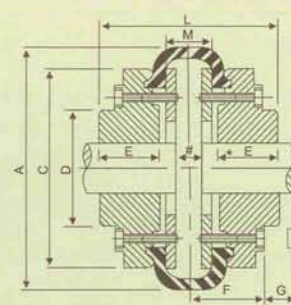
B TYPE



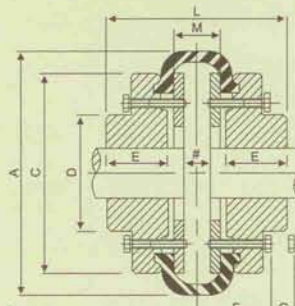
F - 40 & 45



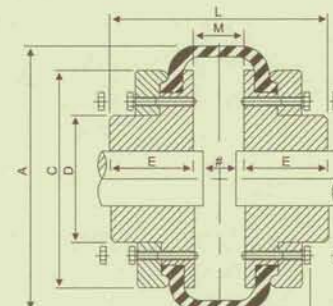
F - 50



F - 60



F - 70 TO 120



F - 140 TO 250

TABLE 2 : DIMENSIONS "F&H" AND "B" TYPE COUPLINGS

Size	TYPE F & H								TYPE B								A	C	‡ G	† M	No. of screws per flange
	Bush No.	Max Bore	L	D	E	F	ξ J	* Approx Weight Kg	Max Bore	Min Bore	L	D	E	F	Set Screw on key	* Approx Weight Kg					
F40	1008	25	67	-	22	33.5	29	1.4	30	11.00	67	-	22	33.5	M5	2.0	104.0	82	43	23	4
F45	1108	28	67	-	22	33.5	29	3.0	32	11.00	73	-	25	36.5	M5	2.2	120.0	94	43	23	4
F50	1210	32	78	79.0	25	39.0	38	3.1	38	16.00	92	79	32	46.0	M5	4.0	133.5	100	43	28	4
F60	1610	42	86	103.0	25	43.0	38	5.2	48	16.00	112	73	38	43.0	M6	5.0	165.0	125	43	36	5
F70	1610	42	92	76.0	25	50.5	38	7.4	55	19.05	132	82	45	50.5	M6	8.0	197.0	144	10	42	5
F80	2012	50	111	95.0	32	53.0	47	9.2	65	25.40	149	95	51	53.0	M10	12.0	211.0	167	10	47	6
F85	2012	50	112	103.0	32	53.5	47	12.5	70	31.75	154	103	53	53.5	M12	14.0	222.0	179	13	48	6
F90	2517	60	140	110.0	45	59.5	50	15.0	76	31.75	164	110	57	59.5	M12	15.0	235.0	188	13	50	6
F100	2517	60	148	124.0	45	61.5	50	20.0	85	31.75	178	124	60	61.5	M12	21.0	254.0	216	13	58	6
F110	2517	60	140	134.0	45	63.5	50	26.5	90	31.75	180	134	65	63.5	M12	28.0	279.0	233	14	50	6
F120	3020	75	157	152.0	51	70.0	68	35.5	102	38.10	207	152	76	70.0	M12	41.0	314.0	264	14	55	6
F140	3535	90	204	194.5	89	76.0	89	67.2	120	75.00	204	195	89	76.0	M20	61.0	359.0	313	14	26	8
F160	4040	100	220	216.0	102	78.0	110	91.0	140	75.00	220	216	102	78.0	M20	86.0	402.0	345	19	16	8
F180	4545	110	258	266.0	114	94.0	126	146.0	150	75.00	258	266	114	94.0	M20	141.0	470.0	398	19	30	10
F200	4545	110	278	266.0	114	103.0	126	182.0	150	75.00	276	266	114	103.0	M20	179.0	508.0	429	19	48	12
F220	5050	125	312	267.0	127	118.0	140	320.0	160	90.00	312	267	127	118.0	M20	312.0	562.0	474	20	56	12
F250	-	-	-	-	-	-	-	-	190	100.00	360	290	150	125.0	M20	500.0	628.0	532	25	60	12

Dimensions are in millimetres

* Weights given are for min. bore complete coupling.

† M is the distance between flanges.

Shaft ends, although normally located 'M' apart - can project beyond the flanges as shown. In this event, allow sufficient space between shaft ends for the float and misalignment.

‡ G is the amount by which clamping screws need to be withdrawn to release tyre.

ξ J is the wrench clearance to allow for tightening and loosening the bush on the shaft. The use of shortened wrench will allow this dimension to be reduced.

FENNER TYRE COUPLINGS - POWER RATINGS



TABLE 3 : POWER RATINGS (kW)

Speed (rev/min)	COUPLING SIZE																
	F40	F45	F50	F60	F70	F80	F85	F90	F100	F110	F120	F140	F160	F180	F200	F220	F250
100	0.22	0.39	0.56	1.11	1.70	2.65	3.2	3.82	5.29	7.46	12.4	19.7	32.6	57.4	84	104	132
200	0.44	0.78	1.11	2.22	3.39	5.30	6.4	7.64	10.00	14.90	24.8	39.4	65.2	115	168	209	264
300	0.66	1.17	1.67	3.33	5.09	7.95	9.6	11.50	15.90	22.40	37.1	59.1	97.8	172	252	313	396
400	0.88	1.56	2.22	4.44	6.79	10.60	12.8	15.30	21.20	29.80	49.5	78.8	130	230	336	418	529
500	1.10	1.95	2.78	5.55	8.48	13.20	16.0	19.10	26.40	37.30	61.9	98.5	163	287	420	522	660
600	1.32	2.34	3.33	6.66	10.20	15.90	19.2	22.90	31.70	44.70	74.3	118	196	345	504	627	793
700	1.54	2.73	3.89	7.77	11.90	18.50	22.4	26.80	37.00	52.20	86.6	138	228	402	588	731	925
720	1.58	2.80	4.00	7.99	12.20	19.10	23.0	27.50	38.10	53.70	89.1	142	235	414	605	753	951
800	1.76	3.12	4.44	8.88	13.60	21.20	25.6	30.60	42.30	59.60	99.0	158	261	459	672	836	1057
900	1.98	3.00	5.00	9.99	15.30	23.80	28.8	34.40	47.60	67.10	111.0	177	293	517	756	940	1198
960	2.11	3.74	5.33	10.70	16.30	25.40	30.7	36.70	50.80	71.60	119.0	189	313	551	806	1003	1269
1000	2.20	3.90	5.55	11.10	17.00	26.50	32.0	38.20	52.90	74.60	124.0	197	326	574	840	1045	1322
1200	2.64	4.68	6.66	13.30	20.40	31.80	38.4	45.90	63.50	89.50	149.0	236	391	689	1008		
1400	3.08	5.46	7.77	15.50	23.80	37.10	44.8	53.50	74.00	104.00	173.0	276	456	804			
1440	3.17	5.61	7.99	16.00	24.40	38.10	46.0	55.00	76.10	107.00	178.0	284	469	827			
1600	3.52	6.24	8.88	17.80	27.10	42.40	51.2	61.20	84.60	119.00	198.0	315	522				
1800	3.96	7.02	9.99	20.00	30.50	47.70	57.6	68.80	95.20	134.00	223.0	355					
2000	4.40	7.80	11.10	22.20	33.90	53.00	64.0	76.40	106.00	149.00	248.0						
2200	4.84	8.58	12.20	24.40	37.30	58.30	70.4	84.10	116.00	164.00							
2400	5.08	9.36	13.30	26.60	40.70	63.60	76.8	91.70	127.00								
2600	5.72	10.14	14.40	28.90	44.10	68.90	83.2	99.40	137.00								
2800	6.16	10.92	15.50	31.10	47.50	74.20	89.6	107.00									
2880	6.33	11.23	16.00	32.00	48.90	76.30	92.1	110.00									
3000	6.60	11.70	16.70	33.30	50.90	79.50	96.0										
3500	7.70	13.65	19.40	38.90	59.40	The figures in heavier type are for standard motor speeds											
3600	7.92	14.04	20.00	40.00													

For speeds below 100 rev/min and intermediate speeds, use normal torque ratings.

FENNER TYRE COUPLINGS - PHYSICAL CHARACTERISTICS



TABLE 4 : PHYSICAL CHARACTERISTICS

Size	Max. Speed (rev/min)	Torque (Nm)		Moment of inertia MR^2 (kgm ²)	Torsional Stiffness (Nm/°)	Maximum Misalignment (mm)	
		Nominal	Max.			Parallel	End float ±
F40	4500	21	64	0.00148	5	1.1	1.3
F45	4500	37	110	0.00250	9	1.2	1.5
F50	4500	53	160	0.00349	13	1.3	1.7
F60	4000	106	318	0.01030	26	1.6	2.0
F70	3600	162	487	0.01811	41	1.9	2.3
F80	3100	253	759	0.03679	63	2.1	2.6
F85	3000	305	915	0.05015	76	2.2	2.8
F90	2880	365	1096	0.06374	91	2.4	3.0
F100	2600	505	1517	0.11989	126	2.6	3.3
F110	2300	712	2137	0.16012	178	2.9	3.7
F120	2050	1182	3547	0.34302	296	3.2	4.0
F140	1800	1881	5642	0.69452	470	3.7	4.6
F160	1600	3113	9339	1.21767	778	4.2	5.3
F180	1500	5485	16455	2.01800	1371	4.8	6.0
F200	1300	8022	23508	4.03446	1959	5.3	6.6
F220	1100	9932	33125	8.67644	2760	5.8	7.3
F250	1000	12606	42740	16.85095	3562	6.6	8.2

Notes :

1. Maximum torque figures should be regarded as short duration overload ratings for use in such circumstances as direct-on-line starting etc.
2. All flexible tyres have an angular misalignment capacity upto 4°

TABLE 5

Coupling Size	F40*	F45*	F50*	F60*	F70	F80	F85	F90	F100	F110	F120	F140	F160	F180	F200	F220	F250
M2 (mm)	23	23	28	36	42	47	48	50	58	50	55	26	16	30	48	56	60
Clamping Screw Torque	Nm	15	15	15	15	24	24	32	32	32	32	35	35	35	35	38	38

* Hexagonal Socket Cap Head Clamping Screws on these sizes.

FENNER TYRE COUPLINGS - INSTALLATION



INSTALLATION INSTRUCTIONS

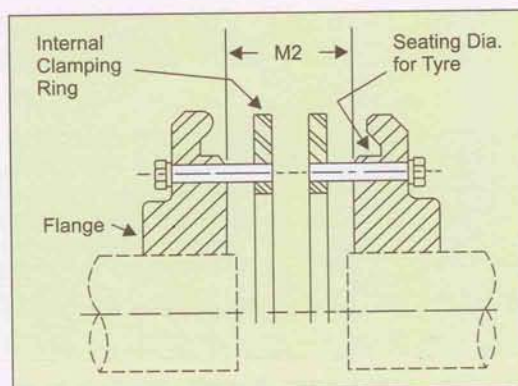
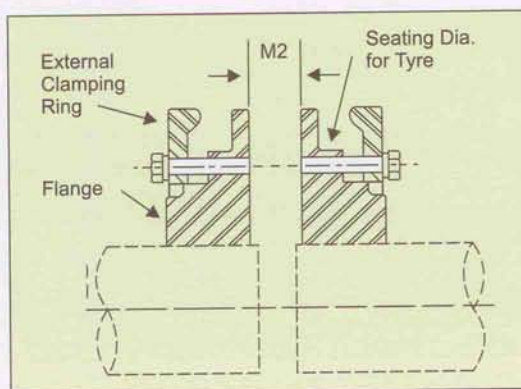
Note : Satisfactory performance depends on correct installation and maintenance. All instructions in this manual must therefore be followed carefully.

1. Thoroughly clean all components, paying particular attention to the removal of the protective coating in the bore of the flanges.
2. Fit flanges to the shafts placing the external clamp rings on the shafts. (Where Taper-Lock® flanges are used, see separate fitting instructions supplied with the Taper-Lock® Bushes). Locate flanges so that dimension M2 is obtained (see paragraph 3). Flanges with internal clamping rings should then have the clamping rings fitted, engaging only two or three of the threads of the screws at this time.
3. Bring shafts into line until dimension M2 is obtained (table 5). If shaft end float is to occur, locate the shafts at mid-position of end float when checking dimension M2. Note that shaft ends may project beyond the faces of the flanges if required. In this event, allow sufficient space between shaft ends for end float and misalignment. Flanges should be fitted flush with the end of the shaft when used with Mill-Motor flanges.
4. Check parallel alignment by laying a straight edge across the flanges at several positions around the circumference. Check angular alignment by measuring gap between flanges at several positions around the circumference. It is desirable to align the coupling as accurately as possible, particularly on high speed applications.
5. Open out tyre and fit over coupling flanges ensuring that the tyre beads seat properly on the flanges and/or clamping rings. To ensure proper seating, it may be necessary to strike the outside diameter of the tyre with a small mallet. When seated, there should be a gap between the ends of the tyre as shown in table 6.

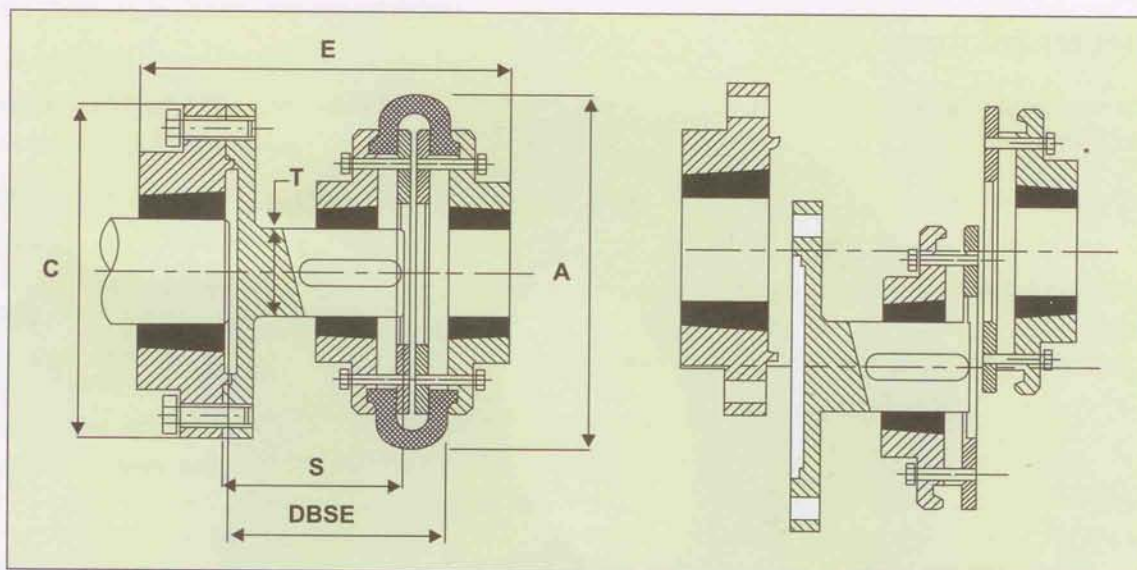
TABLE 6

COUPLING SIZE	F40 TO F60	F70 TO F120	F140	F160 TO F180	F200 TO F250
Tyre Gap in mm.	2	3	5	6.5	8

6. Tighten clamping ring screws alternately and evenly (half turn at a time) working round each flange until the required screw torque is achieved.



FENNER TYRE SPACER COUPLINGS



Comprising a Fenner Tyre coupling (size F40-F140) complete with a spacer flange designed for use on applications where it is an advantage to be able to move either shaft axially without disturbing the driving or driven machine; e.g. centrifugal pump rotors. Fenner Type spacer couplings are primarily designed for the standard distance between shaft end dimensions 100, 140 and 180 mm.

TABLE 7

SPACER SIZE	DBSE	FENNER TYRE COUPLING SIZE	SPACER BUSH SIZE	MAX. BORE	FENNER TYRE COUPLING BUSH SIZE	MAX. BORE	A	C	E	S	T
SM12	80	F40	1210	32	1008	25	104	118	134	77	25
SM12	100	F40	1210	32	1008	25	104	118	140	97	25
SM16	100	F40 *	1615	42	1008	25	104	127	170	94	32
SM16	140	F40 *	1615	42	1008	25	104	127	210	134	32
SM16	100	F50	1615	42	1210	32	133.5	127	173	94	32
SM16	140	F50	1615	42	1210	32	133.5	127	213	134	32
SM16	100	F60	1615	42	1610	42	165	127	177	94	32
SM16	140	F60	1615	42	1610	42	165	127	214	134	32
SM25	100	F70	2517	60	1610	42	197	178	180	94	42
SM25	140	F70	2517	60	1610	42	197	178	220	134	42
SM25	180	F70	2517	60	1610	42	197	178	260	174	42
SM25	100	F80	2517	60	2012	50	211	178	193	94	48
SM25	140	F80	2517	60	2012	50	211	178	233	134	48
SM25	180	F80	2517	60	2012	50	211	178	273	174	48
SM25	140	F90	2517	60	2517	60	235	178	235	134	48
SM25	180	F90	2517	60	2517	60	235	178	275	174	48
SM30	140	F100	3030	75	2517	60	254	216	269.5	134	60
SM30	180	F100	3030	75	2517	60	254	216	309.5	174	60
SM30	140	F110	3030	75	2517	60	279	216	369.5	134	60
SM30	180	F110	3030	75	2517	60	279	216	309.5	174	60
SM35	140	F120	3535	90	3020	75	314	248	297.5	134	75
SM35	180	F120	3535	90	3020	75	314	248	327.5	174	75
SM35	140	F140	3535	90	3535	90	359	248	296	134	80
SM35	180	F140	3535	90	3535	90	359	248	336	174	80

* F40 'B' Flange must be used to fit spacer shaft.



Using Fenner Taper Taper-Lock® Bushes, it is possible for unskilled labour to achieve 'shrink fit' of pulleys, coupling etc., onto shafts using only a hexagonal wrench.

The arrangement of half-threaded holes and longitudinally split tapered bushes ensures maximum grip and fast, easy fitting. Tightening of the screws into the threaded holes in the hub forces the bush into the taper bored components, thereby effectively contracting the bore of the Taper-Lock® Bush until the equivalent of a 'shrink fit' is obtained.

Taper-Lock® Bushes are suitable for metric as well as imperial shafts.

ADVANTAGES

- No reboring and keywaying costs.
- Saves time and cost in fitting.
- Eliminates precision taper fitting keys.
- 239 bush size/bore combinations are available.
- Interchangeable between many products.
- Taper bored components can be transferred to other diameter shafts by fitting alternative bore bushes.
- Convenience in dismantling for maintenance and component replacement.
- Accommodates shaft limits of +0.051 mm/-0.127mm.

The benefits of using Taper-Lock® Bushes can be extended to include components which have a parallel bore by incorporating Taper-Lock® Adaptors, Taper-Lock® Bolt-on Hubs or Taper-Lock® weld-on Hubs.

FENNER TAPERLOCK® BUSHES



TABLE 8

METRIC BORES AND KEYWAYS

Bore Dia.	Keyway		Shallow Keyway Depth	Catalogue Code Group 029 ...																
	Width	Depth		1008	1108	1210	1215	1310	1610	1615	2012	2517	2525	3020	3030	3525	3535	4040	4545	5050
9	3	1.4	-	...009	009															
10	3	1.4	-	...010	010															
11	4	1.8	-	...011	011	011	011													
12	4	1.8	-	...012	012	012	012													
14	5	2.3	-	...014	014	014	014	014	014	014	014									
16	5	2.3	-	...016	016	016	016	016	016	016	016	016								
18	6	2.8	-	...018	018	018	018	018	018	018	018	018								
19	6	2.8	-	...019	019	019	019	019	019	019	019	019	019							
20	6	2.8	-	...020	020	020	020	020	020	020	020	020	020	020						
22	6	2.8	-	...022	022	022	022	022	022	022	022	022	022	022						
24	8	3.3	1.3	...024*	024	024	024	024	024	024	024	024	024							
25	8	3.3	1.3	...025*	025	025	025	025	025	025	025	025	025	025						
28	8	3.3	1.3		028*	028	028	028	028	028	028	028	028	028						
30	8	3.3	-			030	030	030	030	030	030	030	030	030						
32	10	3.3	1.3			032*	032*	032	032	032	032	032	032	032						
35	10	3.3	1.3					035*	035	035	035	035	035	035	035	035	035			
38	10	3.3	-						038	038	038	038	038	038	038	038	038			
40	12	3.3	1.3						040*	040*	040	040	040	040	040	040	040	040		
42	12	3.3	1.3						042*	042*	042	042	042	042	042	042	042	042		
45	14	3.8	-								045	045	045	045	045	045	045	045		
48	14	3.8	-								048	048	048	048	048	048	048	048		
50	14	3.8	2.8								050*	050	050	050	050	050	050	050		
55	16	4.3	-									055	055	055	055	055	055	055	055	
60	18	4.4	-									060	060	060	060	060	060	060	060	
65	18	4.4	-											065	065	065	065	065	065	
70	20	4.9	-											070	070	070	070	070	070	070
75	20	4.9	-											075	075	075	075	075	075	075
80	22	5.4	-													080	080	080	080	080
85	22	5.4	-													085	085	085	085	085
90	25	5.4	3.4													090*	090*	090	090	090
95	25	5.4	-															095	095	095
100	28	6.4	5.4															100*	100	100
105	28	6.4	-																105	105
110	28	6.4	-																110	110
115	32	7.4	-																	115
120	32	7.4	-																	120
125	32	7.4	-																	125
Nominal dia at large end of Taper				35.0	38.0	47.5	47.5	51.0	57.0	57.0	70.0	85.5	85.5	108.0	108.0	127.0	127.0	146.0	162.0	177.5
Approx. Mass of Bush (Kg)				0.1	0.1	0.2	0.3	0.3	0.3	0.5	0.7	1.5	1.9	2.7	3.6	3.8	5.0	7.7	10	14



Dimensions in millimeters

Keyways are British Standard Metric B.S. 4235: Part 1:1972 and conform to I.S.O. recommendations except for the bore sizes marked * which are shallower.

Where a key is to be used it should be parallel and side fitting with top clearance. Depth of keyway is measured at CENTRE.

Note : Taper-Lock® Bushes with imperial bores can also be supplied. Please consult Fenner.

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14/32/2008-5k

Essex Jaw Couplings



Fenner

POWERTRAN



SALIENT FEATURES

- Simple construction - quick easy installation - No special tools required.
- Flexible insert caters for incidental angular, parallel and axial misalignment.
- Absorbs shock loads and damps small amplitude vibration.
- Insert design presets correct distance between hubs, using raised pads on each leg of the insert.
- Available in a range of stock bore sizes. Can also be supplied with finished bore & keyway.
- Unaffected by moisture, grease and oils-including non-aromatic and non-ketone solvents and temperatures within the range - 40°C to + 100°C.
- Spacer coupling with spacer size depending upon the distance between two shaft ends (DBSE).

SELECTION

Details required for couplings selection

1. Type of driven machine and operating hours per day.

2. Speed and power absorbed by driven machine (if absorbed power is not known, calculate on power rating of prime mover).
3. Diameter of shafts to be connected.
4. Distance between two shaft ends in case of spacer coupling.

PROCEDURE

a) Service Factor

Determine the required service factor from table.

b) Design Power

Multiply the normal running power by the service factor. This gives the Design Power which is used as a basis for selecting the coupling.

c) Coupling Size

Depending upon the type of coupling required, refer to respective power rating tables. Power ratings can be interpolated in relation to speed parameters.

d) Bore Size

Check from the dimension table to see if bore capacity of the couplings is adequate. Otherwise select next higher size coupling.

SERVICE FACTORS

Type of Driven Machine	Type of Driving Unit		
	Electric Motors & Steam Turbines	Internal Combustion Engines	
		More than six cylinders	Less than six cylinders
Uniform Load Agitators, Brewing machinery, Centrifugal compressors and pumps, Belt conveyors, Dynamometers, Lineshafts, Fans upto 7.5 kW, Blowers and exhausters except positive displacement, Generators.	1.0	1.5	2.0
Moderate Shock Clay working machinery, General machine tools, Paper mill beaters and winders, Rotary pumps, Rubber extruders, Rotary screens, Textile machinery, Marine propellers and fans over 7.5 kW.	1.5	2.0	2.5
Heavy Shock Bucket elevators, Cooling tower fans, Piston compressors and pumps, Foundry machinery, Metal presses, Paper mill calenders, Hammer mills, Presses and pulp grinders, Rubber calenders, Pulverisers and positive displacement blowers.	2.0	2.5	3.0

Essex Jaw Couplings



STANDARD COUPLINGS

Size	Power per 100 rev/min kW	Bore in mm		Dimensions in mm				
		Min.	Max.	A	B	D	F	G
F - 095	0.21	15	28	63	54	49	2	25
F - 099	0.39	20	30	72	65	51	2	27
F - 0100	0.50	20	38	88	65	57	2	35
F - 0110	0.92	20	42	108	85	76	3	43
F - 0150	1.50	30	48	115	96	80	3	45
F - 0190	2.02	36	55	133	115	102	3	54
F - 0225	2.75	40	60	153	127	108	3	64

STANDARD SPACER COUPLINGS

Size	Power per 100 rev/min kW	Bore in mm		Distance between shaft ends (DBSE) A	Dimensions in mm		
		Min.	Max.		B	C	X
F-095 S	0.21	15	28	90/100	54	25	6
F-0100 S	0.50	20	38	90/100/140	65	30	6
F-0110 S	0.92	20	42	90/100/140	85	35	8
F-0150 S	1.50	30	48	90/100/140	96	45	10
F-0190 S	2.02	36	55	90/100/140	115	51	10
F-0225 S	2.75	40	60	90/100/140	127	57	12

EXTERNAL SPIDER COUPLINGS

Size	Power per 100 rev/min kW	Bore in mm		Dimensions in mm				
		Min.	Max.	A	B	D	F	G
F - 095 E	0.23	15	23	63	64	49	2	25
F - 099 E	0.38	20	30	72	77	51	2	27
F - 0100 E	0.50	20	38	88	77	57	2	35
F - 0110 E	0.91	20	42	108	95	76	3	43
F - 0150 E	1.47	30	48	115	110	80	3	45
F - 0190 E	2.03	36	55	135	128	102	3	54
F - 0225 E	2.80	40	60	153	141	108	3	64

EXTERNAL SPIDER ALUMINIUM SPACER COUPLINGS

Size	Power per 100 rev/min kW	Bore in mm		Distance between shaft ends (DBSE) A	Dimensions in mm			
		Min.	Max.		B	C	D	F
F-095 ES	0.23	15	28	90/100/140	64	25	49	2
F-0100 ES	0.50	20	38	90/100/140	77	35	57	2
F-0110 ES	0.91	20	42	90/100/140	95	43	76	3
F-0150 ES	1.47	30	48	90/100/140	110	45	80	3
F-0190 ES	2.03	36	55	90/100/140	128	54	102	3
F-0225 ES	2.80	40	60	90/100/140	141	64	108	3

CUSHION COUPLINGS

Size	Power per 100 rev/min kW	Bore in mm		Dimensions in mm				
		Min.	Max.	A	B	D	F	G
F-0226 P	3.45	25	65	178	143	115	3	70
F-0276 P	5.60	25	75	200	163	127	3	80
F-0280 P	8.20	30	75	200	200	140	3	80
F-0295 P	13.40	40	90	238	245	160	3	95
F-02955 P	22.40	50	100	264	245	180	3	108

CUSHION SPACER COUPLINGS

Size	Power per 100 rev/min kW	Bore in mm		Distance between shaft ends (DBSE) A	Dimensions in mm			
		Min.	Max.		B	C	D	X
F-0226 PS	3.45	25	65	135/140/180	145	50	134	12
F-0276 PS	5.60	25	75	135/140/180	165	60	130	12
F-0280 PS	8.20	30	75	135/140/180	200	60	130	14
F-0295 PS	13.40	40	90	135/140/180	249	65	160	16
F-02955 PS	22.40	50	90	135/140/180	249	80	160	16

TYPICAL APPLICATIONS

Pumps including back-pull-out type, Conveyors, Elevators, Packaging Machinery, Food Processing Plants, Compressors, General Machine Tools, Blowers, Paper Mill Beaters and Calenders, etc.



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Fenner Bush Type Flexible Couplings



Fenner

POWERTRAN

BUSH TYPE FLEXIBLE COUPLING SELECTION

The function of a flexible coupling is to transmit torque from one shaft to another in cases where limited misalignment may occur and also to absorb shock loads.

The Fenner Bush Type Flexible Coupling of the cushioned drive type, transmits the torque through high tensile steel bolts to the machine input shaft. Highly developed rubber compounds are used in bushes to absorb shock loads, torsional vibrations and slight misalignments.

Simple and compact in construction, the Fenner Coupling is capable of transmitting high torques at maximum speeds. The flanges are manufactured with cast iron, grade 20 of IS. 210. This type of coupling permits drive in either direction and requires neither lubrication nor adjustment after fitting. The flexible bushes remain unaffected by water, dust and atmospheric conditions.

Machines which are to be coupled by flexible couplings should first be aligned with all possible accuracy. The capacity of the coupling will then deal with misalignments which occur by reason of temperature variations or heavy shaft loading. Setting of machine foundations or bearing wear will also cause extra loading to be imposed on the coupling. Any, or all of these conditions can occur once the machines have been coupled.

Flanges are bored to suit requirements and are keywayed to British Standard Specification, unless otherwise stated. They can also be supplied with the listed minimum bore to permit machining on site.

Power requirements for the standard couplings range from 0.81 Kw to 249 Kw at 100 r.p.m. and sizes from BC1 to NBC11

Details required for coupling selection are :

1. Type of driven machine and operating hours per day.
2. Speed and power absorbed by driven machine (If absorbed power is not known, it is calculated based on power rating of the prime mover).
3. Diameter of shafts to be connected.

PROCEDURE :

- a) Service Factor** : Determine the required service factor from Table 1.
- b) Design Power** : Multiply the normal running power by the service factor. This gives the Design Power which is used as a basis for selecting the coupling.
- c) Coupling Size** : Refer to Table 3 and from the appropriate speed, read across until a power greater than that required in step (b) is found. The size of coupling required is given at the head of that column.
- d) Bore Size** : Check from dimension in Table 2, whether the chosen flanges can accommodate the required bores.

TABLE 1 : SERVICE FACTORS	PRIME MOVER				
DRIVEN MACHINE	Electric Motor Steam Turbine Shafting	Steam Engine Water Turbine	IC Engine Multi-Cylinder	IC Engine Single Cylinder Diesel Multi-Cylinder	Diesel Engine Single Cylinder
Even Torque Machines; Smooth Loads, Generators; Centrifugal Pumps; Blowers; Small Fans; Line Shafting.	1.00	1.25	1.50	2.00	2.50
Machine Tools (light); Beaters; Exhausters; Wood-working Machines (light); Alternators; Welding Generators; Textile Machines.	1.25	1.50	1.75	2.25	2.75
Multi-Crank Compressors and Pumps; Generators (fluctuating loads); Rotary Dryers & Screens; Rotary Compressors; Planers; Wood-Working Machines (heavy); Pulp Grinders; Shakers; Mine Fans	1.50	1.75	2.00	2.75	3.00
Wire Mills; Cement Mills; Small Printing Presses.	1.75	2.00	2.25	3.00	3.25
Single Crank Compressors & Pumps; Hammers; Ball & Tube Mills; Rolling Mills (light); Shearing Machines; Punches; Rock& Stone Crushers; Brick Making and similar Machines; Printing Presses (large); Grinders; Pulverisors; Cranes & Winches; Mechanical Shovels & Dredges; Winding Gears and Drums.	2.00	2.25	2.50	3.25	3.50
Heavy Rolling Mill Drives; Continuous, Prolonged & Reversing Drives; Severe Traction and Haulage Loads.	2.25	2.50	2.75	3.50	3.75

BUSH TYPE FLEXIBLE COUPLING SELECTION

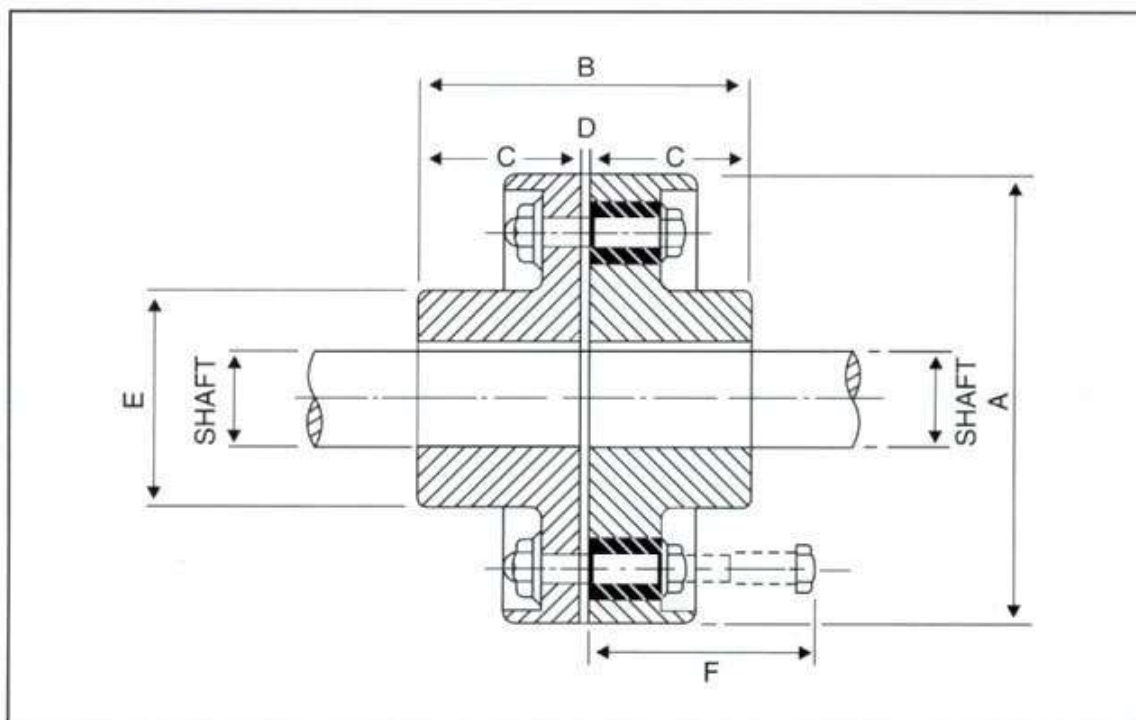


TABLE 2 : DIMENSIONS

Size	No. of Pins	Torque Nm	Min. Bore (mm)	Max. Bore (mm)	Max. Speed rev/min	A	B	C	D	E	F
BC1	3	77	12.7	28	6100	95	79	38	3	40	58
BC2	4	310	12.7	30	5100	114	99	48	3	42	70
BC2A	6	516	16	42	4400	130	105	51	3	60	70
BC3	4	621	16	48	3600	160	107	51	5	68	114
BC4	4	831	20	65	3000	191	125	60	5	90	114
BC4A	6	1241									
BC5	6	1662	25	75	2600	225	157	76	5	105	114
BC6	8	2359	45	95	2300	254	183	89	5	135	114
BC6A	10	2932									
BC6B	12	3533									
NBC7	12	4154	60	115	1950	290	235	115	5	170	114
NBC7A	14	5195	60	120	1900	300	235	115	5	180	130
NBC8	16	5816	65	130	1850	310	255	125	5	195	130
NBC8A	18	7268	65	135	1650	340	265	130	5	200	130
NBC8B	12	8729	70	140	1590	360	276	135	6	210	200
NBC9	13	9932	80	150	1470	390	316	155	6	225	200
NBC9A	15	13274	90	160	1400	410	336	165	6	240	200
NBC10	16	14420	100	170	1300	440	366	180	6	255	200
NBC10A	17	18050	110	180	1200	480	386	190	6	270	212
NBC11	20	23780	120	190	1080	530	406	200	6	285	212

All dimensions are subject to alteration without notice

BUSH TYPE FLEXIBLE COUPLING SELECTION

TABLE 3 : POWER RATINGS (Kw)

Speed rev/min	COUPLING SIZES																			
	BC1	BC2	BC2A	BC3	BC4	BC4A	BC5	BC6	BC6A	BC6B	NBC7	NBC7A	NBC8	NBC8A	NBC8B	NBC9	NBC9A	NBC10	NBC10A	NBC11
100	0.81	3.25	5.4	6.5	8.7	13	17.4	24.7	30.7	37	43.5	54.4	60.9	76.1	91.4	104	139	151	189	249
200	1.62	6.5	10.8	13	17.4	26	34.8	49.4	61.4	74	87	108.8	121.8	152.2	182.8	208	278	302	378	498
300	2.43	9.75	16.2	19.5	26.1	39	52.2	74.1	92.1	111	130.5	163.2	182.7	228.3	274.2	312	417	453	567	747
400	3.24	13	21.6	26	34.8	52	69.6	98.8	122.8	148	174	217.6	243.6	304.4	365.6	416	556	604	756	996
500	4.05	16.25	27	32.5	43.5	65	87	123.5	153.5	185	217.5	272	304.5	380.5	457	520	695	755	945	1245
600	4.86	19.5	32.4	39	52.2	78	104.2	148.2	184.2	222	261	326.4	365.4	456.6	548.4	624	834	906	1134	1494
700	5.67	22.75	37.8	45.5	60.9	91	121.8	172.9	214.9	259	304.5	380.8	426.3	532.7	639.8	728	973	1057	1323	1743
720	5.83	23.4	38.9	46.8	62.6	93.6	125.3	177.8	221	266.4	313.2	391.7	438.5	547.9	658	749	1001	1087	1361	1793
800	6.48	26	43.2	52	69.6	104	139.2	197.6	245.6	296	348	435.2	487.2	608.8	731.2	832	1112	1208	1512	1992
900	7.29	29.25	48.6	58.5	78.3	117	156.6	223.3	276	333	391.5	489.6	548.1	684.9	822.6	936	1251	1359	1701	2241
960	7.77	31.2	51.8	62.4	83.5	124.8	167	237.1	294.7	355.2	417.6	522.2	584.6	730.6	877.4	998	1334	1450	1814	2390
1000	8.1	32.5	54	65	87	130	174	247	307	370	435	544	609	761	914	1040	1390	1510	1890	2490
1200	9.72	39	64.8	78	104.4	156	208.8	296.4	368.4	444	522	652.8	730.8	913.2	1097	1248	1668	1812	2268	
1400	11.34	45.5	75.6	91	121.8	182	243.6	345.8	429.8	518	609	761.6	852.6	1065	1280					
1440	11.66	46.8	77.8	93.6	125.3	187.2	250.6	355.7	442	532.8	626.4	783.4	877	1096	1316					
1600	12.96	52	86.4	104	139.2	208	278.4	395.2	491	592	696	870.4	974.4	1218	1462					
1800	14.58	58.5	97.2	117	156.6	234	313.2	444.6	552.6	666	783	979.2								
2000	16.2	65	108	130	174	260	348	494	614	740										
2200	17.82	71.5	118.8	143	191.4	286	382.8	543.4	675.4	814										
2400	19.44	78	129.6	156	208.8	312	417.6													
2600	21.06	84.5	140.4	169	226.2	338	452.4													
2800	22.68	91	151.2	182	243.6	364														
2880	23.33	93.6	155.5	187.2	250.6	374.4														
3000	24.3	97.5	162	195	261	390														
3500	28.35	113.75	189	260																
4000	32.4	130	216																	
4500	36.45	146.25																		



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